

Short-Term Stock Volatility Regime Modeling

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SIADS 696 – Applied Data Science
University of Michigan
2026

1. Introduction

Financial markets are characterized by time-varying volatility that directly impacts portfolio risk, derivative pricing, and trading strategies. While short-term price prediction remains notoriously difficult due to noise and market efficiency, volatility exhibits persistent structure and clustering that can be exploited for risk-aware decision making. Rather than attempting to forecast asset returns, this project focuses on identifying and predicting **short-term volatility regimes**, defined as discrete market states corresponding to low, medium, and high volatility conditions.

Volatility regime modeling provides a more stable and interpretable alternative to return forecasting. Elevated volatility regimes are typically associated with drawdowns, liquidity stress, and heightened tail risk, while low-volatility regimes often correspond to stable market environments. Accurately identifying these regimes enables investors and risk managers to adjust exposure proactively, improving robustness across market cycles.

This project applies both **supervised and unsupervised learning** methods to volatility regime modeling. Supervised classifiers are trained to predict forward volatility regimes using historical asset-level features and market-wide risk indicators. Unsupervised analysis is used to study regime structure, persistence, and transition dynamics. Compared to related work that focuses primarily on return prediction or single-asset volatility modeling, this project emphasizes interpretable regime behavior across multiple assets using leakage-safe labeling and evaluation.

Our findings show that volatility regimes are both persistent and predictable. Recent realized volatility and VIX-based features dominate predictive performance, while regime transition analysis reveals strong clustering and asymmetric switching behavior consistent with established financial theory.

2. Related Work

Volatility modeling has been extensively studied in both academic and applied finance. Early work on volatility clustering, such as Engle's ARCH and GARCH models, demonstrated that volatility is time-dependent and predictable despite the randomness of returns. These models laid the foundation for modern volatility forecasting.

More recent machine learning approaches have expanded on these ideas. The Kaggle "Two Sigma: Financial Modeling Challenge" explored predicting financial outcomes using engineered features derived from price and volume data. While effective, many such approaches prioritize predictive accuracy over interpretability and often focus on returns rather than volatility regimes.

Other studies have explored regime-based modeling of financial markets, using hidden Markov models or clustering techniques to identify latent market states. These methods provide valuable insights but can suffer from interpretability challenges and sensitivity to assumptions.

This project differs from prior work by explicitly framing volatility prediction as a **classification problem**, combining interpretable feature engineering with regime-based evaluation. Additionally, leakage-safe regime construction and time-aware evaluation ensure that results more closely reflect real-world deployment scenarios.

3. Data Sources

The dataset used in this project consists of daily financial market data obtained from the Yahoo Finance API. Asset-level OHLCV (open, high, low, close, adjusted close, and volume) data were collected for a set of large-cap U.S. equities and exchange-traded funds, including AAPL, MSFT, SPY, QQQ, and IWM. The dataset spans approximately January 2015 through early 2024 at a daily frequency.

Market-wide volatility information was incorporated using the CBOE Volatility Index (VIX), also retrieved via Yahoo Finance. VIX data includes daily closing values and trading volume, providing a proxy for aggregate market risk sentiment.

Raw data were retrieved in Parquet format and processed through a reproducible pipeline. After cleaning, merging, and feature engineering, the final dataset contains approximately **13,600 observations** with **23 features per record**, including asset-level, volume-based, and market-wide volatility indicators.

4. Feature Engineering

Feature engineering focused on capturing short-term volatility dynamics while preventing information leakage. All features were constructed using only information available at or before the prediction time.

Asset prices were transformed into **log returns**, providing a stationary representation of price changes. Rolling **realized volatility** features were computed over 10-day, 20-day, and 60-day windows to capture volatility persistence across multiple horizons.

Trading volume information was incorporated through rolling mean and standard deviation features over 10-day and 20-day windows. These features provide liquidity and market activity signals that may correlate with volatility changes.

Market-wide risk was represented using VIX-based features, including the VIX closing value, rolling realized volatility of the VIX, and VIX log returns. These features allow the models to condition predictions on broader market stress rather than relying solely on asset-specific behavior.

Forward volatility labels were constructed using a 5-day forward realized volatility window. Volatility regimes were defined using quantile thresholds computed **only on the training period**, ensuring

leakage-safe labeling. The final dataset includes a categorical regime label with three classes: low, medium, and high volatility.

5. Part A – Supervised Learning

5.1 Methods Description

Multiple supervised learning models spanning diverse model families were evaluated to predict short-term volatility regimes. This approach allows comparison of linear, probabilistic, and nonlinear methods.

A **logistic regression** classifier was used as a baseline probabilistic model. Logistic regression provides interpretability and serves as a benchmark for assessing whether volatility regimes can be separated using linear decision boundaries. Regularization was applied to mitigate overfitting, and hyperparameters were tuned using cross-validation.

A **Random Forest classifier** was employed as a nonlinear, tree-based ensemble method. Random Forests are well-suited to volatility modeling due to their ability to capture nonlinear relationships and feature interactions. Hyperparameter tuning explored the number of trees, tree depth, and minimum samples per leaf, balancing bias and variance.

All models were trained using temporally ordered train-validation splits to avoid look-ahead bias and better reflect real-world deployment.

5.2 Supervised Evaluation

Evaluation Metrics

Model performance was evaluated using **accuracy** and **macro-averaged F1 score**. Accuracy measures overall classification correctness, while macro F1 ensures balanced evaluation across volatility regimes, preventing dominant regimes from masking poor performance on rarer high-volatility states.

Overall Results

The Random Forest classifier consistently outperformed logistic regression across evaluation metrics. Performance improvements were particularly notable in identifying elevated volatility regimes, which are most critical for risk management applications.

A summary comparison of supervised models is presented in Table 1.

Feature Importance Analysis

Figure 1 shows feature importance derived from the Random Forest model. Recent realized volatility measures (rv_10 and rv_20) dominate predictive performance, followed closely by VIX-based features. This confirms that short-term volatility regimes are primarily driven by volatility persistence and macro-level risk conditions.

Volume-based features contribute additional signal but are consistently less influential, suggesting that liquidity effects are secondary to volatility dynamics.

(Insert Figure 1: Random Forest Feature Importance)

Tradeoffs and Sensitivity

The Random Forest model exhibits tradeoffs between precision and recall across regimes. High-volatility regimes are predicted conservatively, yielding higher precision but lower recall. Sensitivity analysis shows performance is robust to moderate changes in rolling window sizes but degrades significantly when VIX features are removed.

Failure Analysis

Prediction failures frequently occur during rapid regime transitions, when rolling volatility estimates lag true market conditions. Asset-specific shocks not reflected in market-wide indicators also contribute to misclassification. Future improvements could include intraday data or adaptive window sizes.

6. Part B – Unsupervised Learning

6.1 Methods Description

Unsupervised analysis was conducted to study regime structure independent of supervised predictions. Regimes were defined using quantiles of forward realized volatility computed on training data only, ensuring interpretability and leakage safety.

6.2 Unsupervised Evaluation

Figure 2 shows the distribution of volatility regimes, with medium volatility dominating and high volatility occurring less frequently.

(Insert Figure 2: Regime Distribution)

Figures 3 and 4 visualize volatility behavior over time, revealing strong clustering and persistence, particularly during periods of market stress.

(Insert Figure 3: Volatility Over Time)

(Insert Figure 4: Volatility Colored by Regime)

Figure 5 presents the regime transition matrix, showing strong diagonal dominance and frequent transitions between adjacent regimes, consistent with volatility clustering.

(Insert Figure 5: Regime Transition Heatmap)

7. Discussion

Supervised learning results demonstrate that volatility regimes are meaningfully predictable using historical volatility and market-wide risk indicators. Tree-based models effectively capture nonlinear relationships that linear models cannot.

Unsupervised analysis provides valuable interpretability, revealing persistent regimes and structured transitions. Together, supervised and unsupervised approaches offer complementary insights into market risk dynamics.

8. Ethical Considerations

Automated volatility regime models may influence trading and risk decisions. Overreliance on model outputs could amplify market instability or disadvantage less sophisticated participants. These tools should be used as decision-support systems with human oversight and transparency.

9. Statement of Work

Tameem Syed led exploratory data analysis, regime visualization, transition analysis, feature importance interpretation, and contributed to supervised and unsupervised evaluation sections.

Tanzim Chowdhury led data pipeline development, feature engineering, model training, and baseline evaluation.

Both authors collaborated on project design, evaluation methodology, and report writing.

References

Engle, R. F. (1982). Autoregressive Conditional Heteroskedasticity with Estimates of the Variance of UK Inflation. *Econometrica*.
Yahoo Finance API.

CBOE Volatility Index (VIX).
Kaggle Two Sigma Financial Modeling Challenge.